

TrackB Configurable Options Reference

This document lists **all configurable elements** in TrackB that you can modify for different training and evaluation experiments.

📁 Config Files

Located in `local_extraction/configs/`:

Config File	Purpose
<code>trackB.yaml</code>	Main TrackB config (default)
<code>trackB_resnet18_baseline.yaml</code>	ResNet18 baseline preset
<code>trackB_videomae_ego.yaml</code>	VideoMAE egocentric preset
<code>trackC.yaml</code>	TrackC pruning config
<code>base.yaml</code>	Shared base settings

Usage:

```
python -m trackB.trackB_train_loader --config trackB
python -m trackB.trackB_train_loader --config trackB_videomae_ego
```

☑ Compatibility Guide (Backbone / Tokens / Encoder)

Some Track B config keys are **free knobs** (change them anytime), while others are **compatibility knobs** that must match your backbone, pre-extracted tokens, and/or encoder weights.

This section explains how to tell which is which.

How to identify “safe vs compatibility” variables

Use this rule of thumb:

- If a key controls **optimization / losses / evaluation toggles** (e.g., `training.*`, `multi_task.*`, `evaluation.*`), it is usually a **safe knob**.
- If a key controls **tensor shapes / encoder construction / token formats** (e.g., `model.tokenizer.*`, `model.projector.in_dim`, `data.tokens_root`), it is a **compatibility knob**.

Practical method:

1. Start from a known-good preset:
 - ResNet baseline: `--config trackB_resnet18_baseline`
 - VideoMAE: `--config trackB_videomae_ego`
2. Only change **safe knobs** first.
3. If you change a compatibility knob, do a quick smoke run:
 - `python local_extraction/trackB/trackB_train_loader.py --config <preset> --demo --epochs 1`
 - `python local_extraction/trackB/trackB_eval.py --config <preset> --checkpoint <path>`

Safe knobs (generally compatible across backbones)

You can change these without worrying about encoder/token compatibility:

- Training: `training.epochs`, `training.batch_size`, `training.lr`, `training.min_lr`, `training.warmup_epochs`, `training.label_smoothing`, `training.eval_every`, `training.early_stopping.*`, `training.amp`
- Data limits: `training.candidate_limit`, `training.normalize_ttc`
- Multi-task: `multi_task.enabled`, `multi_task.ttc_mode`, `multi_task.ttc_thresholds`, `multi_task.loss_weights.*`
- Eval toggles (usually): hotspot / CLIP flags and weights under `evaluation.*` (these affect scoring, not shapes)

Compatibility knobs (must match the backbone / tokens)

These keys can cause hard errors or silent mismatch if set inconsistently:

1. **Backbone selection**

- `model.tokenizer.video_backbone`
 - `"resnet18"`: 2D per-frame tokens
 - `"videomae_ego"`: 3D VideoMAE tokens
2. **Pre-extracted tokens**
- `data.tokens_root`
 - If set: the tokenizer will load tokens from disk.
 - The tokens in that folder must match the chosen `video_backbone` (and expected token shapes).
 - If the folder is missing or malformed, Track B falls back to on-the-fly extraction (slower).
3. **Projector input dimension (shape-critical)**
- `model.projector.in_dim`
 - Must match the token feature dimension produced by the backbone/tokens:
 - ResNet18 tokens are typically 512
 - VideoMAE base is typically 768
 - If mismatched, you'll get a shape error when projecting tokens.
4. **VideoMAE encoder construction (VideoMAE-only)** Only relevant when `video_backbone: "videomae_ego"`:
- `model.tokenizer.videomae.weights_path` (must exist, otherwise results degrade)
 - `model.tokenizer.videomae.embed_dim, depth, num_heads, patch_size, tubelet_size, num_frames`
 - These must match how the encoder checkpoint was trained.
 - If they do not match, loading weights can fail or produce incompatible outputs.
5. **Temporal window parameters**
- `model.tokenizer.time_len, model.tokenizer.time_stride`
 - These affect what frame window is sampled.
 - For VideoMAE, `time_len` typically must match the pretrained model's expectation (often 16).

Recommended workflow

- If you're using **ResNet18**:
 - Prefer changing: `training.*, multi_task.*, evaluation.*, training.candidate_limit`
 - Be cautious changing: `model.projector.in_dim` (should stay 512)
- If you're using **VideoMAE**:
 - Start from [local_extraction/configs/trackB_videomae_ego.yaml](#)
 - Prefer changing: `training.*, multi_task.*, training.candidate_limit`
 - Avoid changing (unless you know your encoder checkpoint): `videomae.*` block and `projector.in_dim`

Quick “did I break compatibility?” signals

- Immediate runtime error mentioning shapes (e.g., `mat1 and mat2 shapes cannot be multiplied`) → `model.projector.in_dim` mismatch.
- Errors loading model weights → `videomae.*` parameters don't match the checkpoint.
- Eval/training runs but metrics collapse unexpectedly → often wrong `weights_path` (random init) or tokens from a different backbone.

 Model Architecture

Backbone / Tokenizer

Parameter	Values	Default	Description
<code>model.tokenizer.video_backbone</code>	<code>"resnet18", "videomae_ego"</code>	<code>"resnet18"</code>	Feature extraction backbone
<code>model.tokenizer.pretrained</code>	<code>true/false</code>	<code>true</code>	Use pretrained weights
<code>model.tokenizer.freeze_backbone</code>	<code>true/false</code>	<code>true</code>	Freeze backbone weights
<code>model.tokenizer.img_size</code>	<code>int</code>	<code>224</code>	Input image size
<code>model.tokenizer.time_len</code>	<code>int</code>	<code>8 (resnet), 16 (videomae)</code>	Frames in temporal window
<code>model.tokenizer.time_stride</code>	<code>int</code>	<code>2</code>	Step between frames

VideoMAE-Specific

Parameter	Values	Default	Description
<code>model.tokenizer.videomae.hf_model_name</code>	HuggingFace model name	"MCG-NJU/videomae-base"	HF pretrained model
<code>model.tokenizer.videomae.weights_path</code>	path	null	Custom weights (overrides HF)
<code>model.tokenizer.videomae.embed_dim</code>	768/1024/1280	768	Embedding dimension
<code>model.tokenizer.videomae.patch_size</code>	int	16	Spatial patch size
<code>model.tokenizer.videomae.tubelet_size</code>	int	2	Temporal patch size
<code>model.tokenizer.videomae.depth</code>	int	12	Transformer depth
<code>model.tokenizer.videomae.num_heads</code>	int	12	Attention heads
<code>model.tokenizer.videomae.freeze_encoder</code>	true/false	true	Freeze encoder

Projector

Parameter	Values	Default	Description
<code>model.projector.in_dim</code>	int	512 (resnet), 768 (videomae)	Input dimension
<code>model.projector.out_dim</code>	int	256	Output token dimension

Fusion Module (FGTP)

Parameter	Values	Default	Description
<code>model.fusion.dim</code>	int	256	Token dimension
<code>model.fusion.layers</code>	int	2	Number of fusion layers
<code>model.fusion.heads</code>	int	8	Attention heads
<code>model.fusion.dropout</code>	float	0.1	Dropout rate

Prediction Head

Parameter	Values	Default	Description
<code>model.head.dim</code>	int	256	Input dimension
<code>model.head.hidden</code>	int	256	Hidden layer size
<code>model.head.num_classes</code>	int	2	Binary next-active classes
<code>model.head.dropout</code>	float	0.1	Dropout rate

🤖 Training Configuration

Hyperparameters

Parameter	Values	Default	Description
<code>training.epochs</code>	int	20	Number of training epochs
<code>training.batch_size</code>	int	8	Batch size
<code>training.lr</code>	float	1e-3	Learning rate
<code>training.min_lr</code>	float	1e-5	Minimum LR (for scheduler)
<code>training.warmup_epochs</code>	float	1.0	Warmup period

Regularization

Parameter	Values	Default	Description
<code>training.label_smoothing</code>	float	0.05	Label smoothing factor

Parameter	Values	Default	Description
<code>training.weight_decay</code>	float	<code>0.01</code>	Weight decay

Data

Parameter	Values	Default	Description
<code>training.candidate_limit</code>	int	<code>16</code>	Max candidates per frame
<code>training.normalize_ttc</code>	true/false	<code>true</code>	Normalize TTC to [0,1]

Checkpointing

Parameter	Values	Default	Description
<code>training.save_epoch_checkpoints</code>	true/false	<code>true</code>	Save each epoch
<code>training.save_best_checkpoint</code>	true/false	<code>true</code>	Save best model

Early Stopping

Parameter	Values	Default	Description
<code>training.early_stopping.enabled</code>	true/false	<code>true</code>	Enable early stopping
<code>training.early_stopping.patience</code>	int	<code>5</code>	Epochs without improvement
<code>training.early_stopping.monitor</code>	"mAP", "accuracy", "ttc_mae"	<code>"mAP"</code>	Metric to monitor
<code>training.early_stopping.mode</code>	"max", "min"	<code>"max"</code>	Optimization direction

Evaluation During Training

Parameter	Values	Default	Description
<code>training.eval_every</code>	int	<code>1</code>	Evaluate every N epochs

Mixed Precision

Parameter	Values	Default	Description
<code>training.amp</code>	true/false	<code>false</code>	Automatic mixed precision

🌀 Multi-Task Learning

Parameter	Values	Default	Description
<code>multi_task.enabled</code>	true/false	<code>true</code>	Enable multi-task heads
<code>multi_task.predict_noun</code>	true/false	<code>true</code>	Predict noun class
<code>multi_task.predict_verb</code>	true/false	<code>true</code>	Predict verb class
<code>multi_task.predict_ttc</code>	true/false	<code>true</code>	Predict time-to-contact
<code>multi_task.ttc_mode</code>	"reg", "bin"	<code>"reg"</code>	TTC as regression or classification
<code>multi_task.ttc_bins</code>	int	<code>5</code>	Number of TTC bins (if <code>bin</code> mode)
<code>multi_task.ttc_thresholds</code>	list	<code>[0.5, 1.0, 2.0]</code>	TTC bin boundaries

Loss Weights

Parameter	Values	Default	Description
<code>multi_task.loss_weights.next_active</code>	float	<code>1.0</code>	Binary next-active loss weight
<code>multi_task.loss_weights.noun</code>	float	<code>1.0</code>	Noun classification loss weight
<code>multi_task.loss_weights.verb</code>	float	<code>1.0</code>	Verb classification loss weight

Parameter	Values	Default	Description
<code>multi_task.loss_weights.ttc</code>	float	<code>1.0</code>	TTC loss weight

Evaluation Configuration

Basic Settings

Parameter	Values	Default	Description
<code>evaluation.batch_size</code>	int	<code>8</code>	Eval batch size
<code>evaluation.compute_map</code>	<code>true/false</code>	<code>true</code>	Compute mean AP
<code>evaluation.compute_ttc_mae</code>	<code>true/false</code>	<code>true</code>	Compute TTC MAE

Hotspot Priors (PEAR-style)

Parameter	Values	Default	Description
<code>evaluation.hotspot_priors.enabled</code>	<code>true/false</code>	<code>true</code>	Use hotspot priors
<code>evaluation.hotspot_priors.path</code>	path	<code>local_extraction/v2/hotspot_priors_train_logodds_min10.json</code>	Priors file
<code>evaluation.hotspot_priors.alpha</code>	float	<code>0.3</code>	Prior blending weight

CLIP Re-ranking

Parameter	Values	Default	Description
<code>evaluation.clip_rerank.enabled</code>	<code>true/false</code>	<code>true</code>	Use CLIP re-ranking
<code>evaluation.clip_rerank.model</code>	<code>"ViT-B/32", "ViT-L/14", etc.</code>	<code>"ViT-B/32"</code>	CLIP model variant
<code>evaluation.clip_rerank.weight</code>	float	<code>0.3</code>	CLIP score blending weight

Visualization

Parameter	Values	Default	Description
<code>evaluation.save_overlays</code>	<code>true/false</code>	<code>true</code>	Save visualization overlays
<code>evaluation.topk_overlay</code>	int	<code>3</code>	Number of top predictions to visualize

Data Sources

Parameter	Values	Default	Description
<code>data.stageB_run</code>	path or <code>null</code>	<code>null</code> (auto-detect)	Stage B run directory
<code>data.train_manifest</code>	path or <code>null</code>	<code>null</code> (auto-detect)	Training manifest
<code>data.val_manifest</code>	path or <code>null</code>	<code>null</code> (auto-detect)	Validation manifest
<code>data.tokens_root</code>	path or <code>null</code>	<code>null</code>	Pre-extracted tokens directory

TrackC: RGTP Pruning

Parameter	Values	Default	Description
<code>rgtp.enabled</code>	<code>true/false</code>	<code>true</code>	Enable RGTP pruning
<code>rgtp.rate</code>	float (0.0-0.95)	<code>0.1</code>	Fraction of tokens to prune
<code>rgtp.min_keep</code>	int	<code>2</code>	Minimum tokens to keep
<code>rgtp.temporal_decay</code>	float	<code>0.6</code>	Rollout vs motion blend factor

Parameter	Values	Default	Description
rgtp.logit_fill	float	-12.0	Logit value for pruned tokens

Instrumentation

Parameter	Values	Default	Description
instrumentation.record_latency	true/false	true	Measure inference latency
instrumentation.record_vram	true/false	true	Record VRAM usage
instrumentation.record_flops	true/false	true	Profile FLOPs

 CLI Overrides

Training script supports these CLI arguments:

```
python -m trackB.trackB_train_loader \  
  --config trackB           # Config preset  
  --epochs 30               # Override epochs  
  --batch_size 16           # Override batch size  
  --lr 0.0005               # Override learning rate  
  --demo                    # Quick demo mode
```

Evaluation script supports:

```
python -m trackB.trackB_eval \  
  --checkpoint path/to/checkpoint.pt
```

TrackC pruning:

```
python -m trackC.trackC_pruning \  
  --config trackC           # Config preset  
  --checkpoint path/to/checkpoint.pt  
  --rgtp_rate 0.3           # Override pruning rate  
  --no_pruning               # Disable pruning (baseline)  
  --video_backbone videomae_ego
```

 Recommended Ablation Experiments

1. Backbone Comparison

```
# ResNet18 baseline  
python -m trackB.trackB_train_loader --config trackB_resnet18_baseline  
  
# VideoMAE egocentric  
python -m trackB.trackB_train_loader --config trackB_videomae_ego
```

2. Learning Rate Sweep

```
python -m trackB.trackB_train_loader --lr 0.0001  
python -m trackB.trackB_train_loader --lr 0.0005  
python -m trackB.trackB_train_loader --lr 0.001  
python -m trackB.trackB_train_loader --lr 0.005
```

3. Multi-Task Ablation

Edit `trackB.yaml`:

```
multi_task:
  enabled: false # Disable all auxiliary tasks
```

4. Loss Weight Tuning

Edit `trackB.yaml`:

```
multi_task:
  loss_weights:
    next_active: 2.0 # Increase primary task weight
    noun: 0.5
    verb: 0.5
    ttc: 0.5
```

5. Pruning Rate Sweep (TrackC)

```
python -m trackC.trackC_pruning --rgtp_rate 0.0 # Baseline
python -m trackC.trackC_pruning --rgtp_rate 0.1
python -m trackC.trackC_pruning --rgtp_rate 0.3
python -m trackC.trackC_pruning --rgtp_rate 0.5
```

Your Best ResNet18 Config

From `trackB_best.pt` checkpoint:

```
training:
  epochs: 20
  batch_size: 8
  lr: 0.001
  min_lr: 1e-05
  warmup_epochs: 1.0
  label_smoothing: 0.05
  candidate_limit: 16
  normalize_ttc: true
  amp: false
  early_stopping:
    patience: 5
    monitor: "mAP"
    mode: "max"

multi_task:
  enabled: true
  ttc_mode: "reg" # regression
  loss_weights:
    next_active: 1.0
    noun: 1.0
    verb: 1.0
    ttc: 1.0

model:
  tokenizer:
    video_backbone: "resnet18"
```

Result: mAP 35.80%